

A theoretical analysis of scattering loss from planar optical waveguides

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Received 12 October; revised 24 November; accepted 26 November 1993

We derive analytical expressions for the scattering loss in planar optical waveguides whose surface roughness can be described by an exponential or Gaussian autocorrelation function. Our results show, for the first time, the explicit dependence of the loss coefficient on all the waveguide parameters. In addition, we show that for any waveguide there is a simple upper limit for the scattering loss.

1. Introduction

The purpose of this paper is to derive an analytical expression for the radiation loss from a planar optical waveguide. We consider the important problem of radiation loss arising from scattering by surface roughness of the waveguide walls. With recent advances in material technology, the absorption losses of silica-based waveguides have been dramatically reduced so that transmission losses are largely the result of scattering by surface irregularities of the waveguide walls. The original theoretical treatment of the waveguide scattering problem was presented in a paper by Marcuse, in which he modelled the scattering loss as arising from the coupling between the fundamental guided mode of the waveguide and the radiation modes [1]. Marcuse's final expression for the waveguide loss involves a complicated integral over the radiation modes. The complexity of the final result obscures much of the underlying physics, and makes it difficult to anticipate the general effects on waveguide loss of varying the various waveguide parameters; these include the refractive indices, the optical wavelength, and the waveguide thickness, amongst others. These difficulties are exacerbated when one realizes that variations of the waveguide refractive index, for example, can produce either an increase or a reduction in the loss depending on the size of the correlation length of the surface roughness.

The complexity of Marcuse's result stems from the fact that it includes a complete description of both near and far fields; it is, of course, only the latter that is important for determining radiation loss. Consequently, we recently formulated a simpler, but no less accurate analysis of scattering loss [2]. Our method was based on the direct computation of the far field, using the method of equivalent currents; a general description of this approach may be found in Snyder and Love [3]. Using this method we derived the following expression for the exponential radiation loss coefficient for scattering by surface roughness in a symmetric single-

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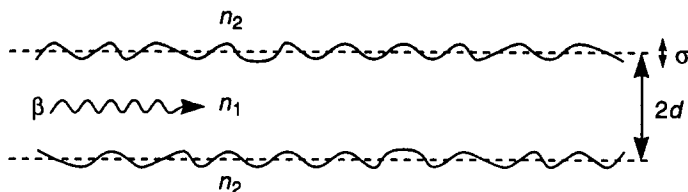


Figure 1 Geometry of the planar waveguide used for the derivation of Equation 1. The surface roughness is described by the r.m.s. deviation σ from flatness of an unperturbed waveguide of width $2d$. The incident TE mode has a propagation constant β .

mode waveguide [2]:

$$\alpha = \varphi^2(d)(n_1^2 - n_2^2)^2 \frac{k_0^3}{4\pi n_1} \int_0^\pi \tilde{R}(\beta - n_2 k_0 \cos \theta) d\theta \quad (1)$$

The waveguide geometry is shown in Fig. 1. In deriving Equation 1 we assumed that scattering occurs independently at both walls of the waveguide. If only one wall is rough then Equation 1 must be multiplied by 1/2, which then corresponds to the situation considered by Marcuse [1]. Equation 1 was also derived independently by Love and Ladouceur using similar methods to our own [4]. In Equation 1, $\varphi(d)$ is the modal field evaluated at the waveguide surface; n_1 and n_2 are the core and cladding refractive indices, respectively; k_0 is the free-space wavenumber and β is the modal propagation constant; the thickness of the waveguide is $2d$. The surface roughness of the waveguide walls is described by the spectral density function $\tilde{R}(\Omega)$, which is related to the autocorrelation function (ACF) $R(u)$ of the surface roughness through the Fourier transform

$$\tilde{R}(\Omega) = \int_{-\infty}^{\infty} R(u) \exp(i\Omega u) du \quad (2)$$

The surface roughness is characterized by a correlation length L_c , and mean square deviation σ^2 from a flat surface. The parameter σ^2 is related to the autocorrelation function $R(u)$ by

$$\sigma^2 = R(0) \quad (3)$$

The modal field $\varphi(y)$ used in Equation 1 is normalized so that

$$\int_{-\infty}^{\infty} \varphi^2(y) dy = 1 \quad (4)$$

Numerical evaluation, using a variety of autocorrelation functions R , shows that Equation 1 agrees very closely with the results of Marcuse (allowing for the factor of 1/2 discussed above) [1, 2]. However, it is apparent that Equation 1 still does not go sufficiently far with regard to providing physical insight for the best waveguide design. There are many parameters that can be varied during waveguide design, namely n_1 , n_2 , d , k_0 , σ , and L_c . Of course, some of these are more under the control of the waveguide designer than others. For example, the wavelength will probably be fixed by system requirements, and n_1 , n_2 , and d will be restricted by the needs of single-mode operation. The details of the surface roughness will depend on the fabrication process. Furthermore, Equation 1 does little to reveal to what extent α can be expressed in terms of normalized waveguide parameters such as U , V , and W or dimensionless

combinations of σ , d and L_c , where [3]

$$U = d\sqrt{n_1^2 k_0^2 - \beta^2} \quad V = k_0 d \sqrt{n_1^2 - n_2^2} \quad \text{and} \quad W = d\sqrt{\beta^2 - n_2^2 k_0^2}$$

Such a solution of Equation 1 is clearly of some importance, since it would greatly simplify waveguide designs aimed at minimizing scattering loss. To achieve this we are therefore forced to evaluate Equation 1 analytically, and this requires an explicit choice to be made for the autocorrelation function $R(u)$. For this we have chosen the commonly used exponential and Gaussian autocorrelation functions:

$$R(u) = \sigma^2 \exp\left(-\frac{|u|}{L_c}\right) \quad (5)$$

and

$$R(u) = \sigma^2 \exp\left(-\frac{u^2}{L_c^2}\right)$$

Recent measurements of silica waveguide surface roughness made with an atomic force microscope have demonstrated that an exponential ACF is a realistic model for the surface roughness statistics of a variety of waveguides [5].

In Sections 2 and 3 we discuss the evaluation of Equation 1, first with an exponential ACF, and then in Section 3 using a Gaussian ACF. In the former case an exact analytical evaluation of Equation 1 is possible. For the Gaussian ACF we have derived an approximate, but very accurate, analytical result by use of the method of steepest descent [6]. Some of the detailed mathematics this involves is presented in an appendix. In both cases, we also derive rigorous upper bounds on the scattering loss that should be of use in deciding whether a particular class of waveguide design is likely to result in acceptable losses.

Finally, in Section 4 we present our conclusions.

2. Exponential autocorrelation function

We now proceed to the evaluation of Equation 1 using the exponential form of the ACF. We require the following integral:

$$S = \int_0^\pi \tilde{R}(\beta - n_2 k_0 \cos \theta) d\theta \quad (6)$$

Inserting the exponential form of $R(u)$ from Equation 5 and using Equation 2, we have

$$S = 2\sigma^2 L_c \int_0^\pi \frac{d\theta}{1 + L_c^2 (\beta - n_2 k_0 \cos \theta)^2} \quad (7)$$

The integral in Equation 7 is straightforward and can be computed by means of the half-angle substitution $t = \tan(\theta/2)$. After some lengthy algebra the final result is

$$S = \sqrt{2}\sigma^2 L_c \pi \frac{\left[\{4\beta^2 L_c^2 + [1 - L_c^2 (\beta^2 - n_2^2 k_0^2)]^2\}^{1/2} + 1 - L_c^2 (\beta^2 - n_2^2 k_0^2) \right]^{1/2}}{\{4\beta^2 L_c^2 + [1 - L_c^2 (\beta^2 - n_2^2 k_0^2)]^2\}^{1/2}} \quad (8)$$

To complete the expression for α we need $\varphi^2(d)$. This is easily obtained from the standard

solution for the TE₀ mode of a planar waveguide [7]:

$$\varphi^2(d) = \frac{2V(1-b)}{d \left[V + \frac{1}{\sqrt{b}} \right]} \quad (9)$$

where V was defined above, and the normalized propagation constant b is defined by

$$b = \frac{\beta^2 - n_2^2 k_0^2}{(n_1^2 - n_2^2) k_0^2}$$

The final expression for α is now obtained by combining Equations 8 and 9. It will prove convenient to introduce the following dimensionless parameters:

$$\Delta = \frac{n_1^2 - n_2^2}{2n_1^2} \quad x = W \frac{L_c}{d} \quad \gamma = \frac{n_2 V}{n_1 W \sqrt{\Delta}}$$

The parameter x represents a normalized correlation length, and γ measures how weakly guiding the waveguide is. The remaining parameters were defined previously. After some algebra our final expression for α is

$$\alpha = \frac{\sigma^2}{\sqrt{2} k_0 d^4 n_1} g(V) f_e(x, \gamma) \quad (10)$$

where

$$g(V) = \frac{U^2 V^2}{(1+W)} \quad (11)$$

and where $f_e(x, \gamma)$ is defined by

$$f_e(x, \gamma) = \frac{x \{ [(1+x^2)^2 + 2x^2\gamma^2]^{1/2} + 1 - x^2 \}^{1/2}}{[(1+x^2)^2 + 2x^2\gamma^2]^{1/2}} \quad (12)$$

Equations 10 to 12 represent a complete analytical solution for scattering loss when the surface roughness is described by an exponential autocorrelation function. To our knowledge, this is the first closed-form solution to this problem. Equation 10 shows, in a very clear way, how α can be expressed in terms of normalized parameters. It also reveals the separate contributions from waveguide geometry and surface roughness. The function $g(V)$ is determined purely by the waveguide geometry, whilst the function $f_e(x, \gamma)$ describes the integral over the spectral density function.

In Fig. 2 we have plotted $g(V)$ and W as functions of V , and in Fig. 3 we have plotted $f_e(x, \gamma)$ as a function of x for various values of γ . Figures 2 and 3 are a set of universal curves which, together with Equation 10, allows the loss of any waveguide to be estimated.

In any practical waveguide the range of V values over which the waveguide will be operated must necessarily be small. On the one hand, V must be less than $\pi/2$ for single-mode operation; on the other hand, V must be greater than about 1 to ensure a reasonably well-confined optical field. As a result we see from Fig. 2 that we do not expect $g(V)$ to lie outside a range from about 0.4 to 0.8 for most practical waveguides. Consequently, scattering loss will be relatively insensitive to changes in $g(V)$ as the waveguide design is varied. However, looking at Fig. 3, much larger variations in $f_e(x, \gamma)$ are expected amongst different wave-

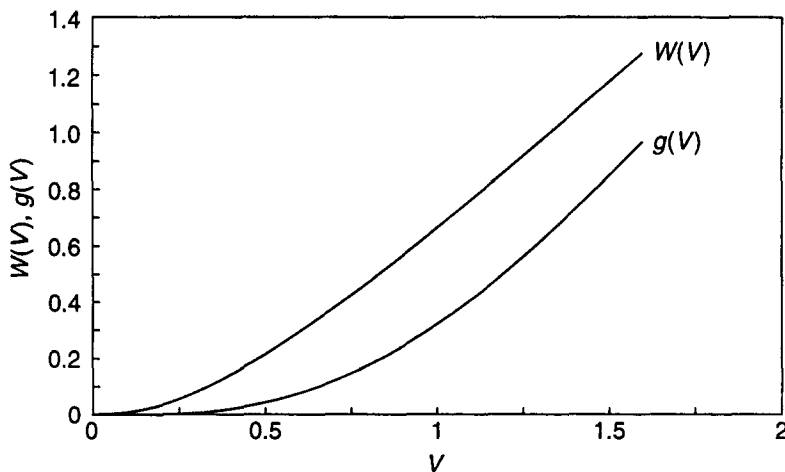


Figure 2 The normalized waveguide parameter W , and the function $g(V)$ used in Equation 11 plotted against V .

guides, and we expect $f_e(x, \gamma)$ to have the largest influence in determining waveguide loss. With reference to Fig. 3, it is then clear that the scattering loss coefficient α always has a maximum value, determined by the turning points of $f_e(x, \gamma)$. We can categorize waveguides as being either weakly guiding, for which $\gamma \gg 1$, or strongly guiding, in which case $\gamma \approx 1$. Of course, most optical waveguides will fall into the former category. For $\gamma \gg 1$ it can be shown, after some lengthy algebra, that the maximum of $f_e(x, \gamma)$ approaches $1/2$, i.e.

$$f_{e,\max} \xrightarrow{\gamma \gg 1} \frac{1}{2} \quad (13)$$

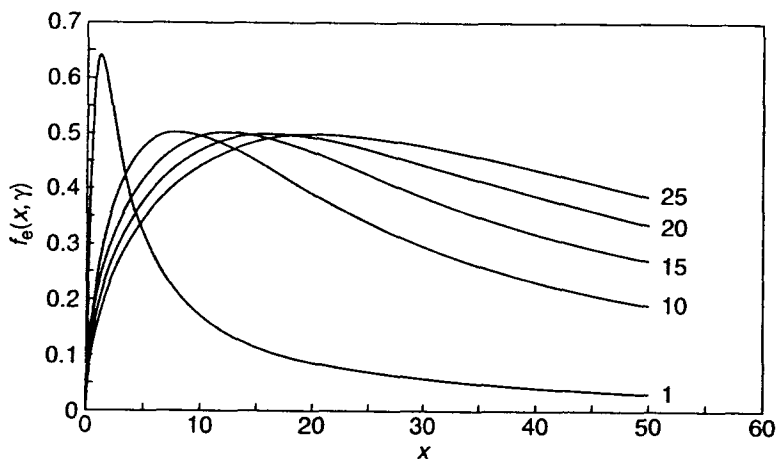


Figure 3 The spectral function $f_e(x, \gamma)$ for an exponential autocorrelation function plotted against x for various values of γ .

with the maximum occurring at

$$x \xrightarrow{\gamma \gg 1} \gamma \sqrt{\frac{2}{3}} \quad (14)$$

Equation 14 has a simple and not unexpected interpretation. If we insert the definitions of x and γ given above, and assume weak guidance, then we find that Equation 14 can be written as

$$L_c = \frac{1}{\sqrt{3}(\beta - n_2 k_0)} \quad (15)$$

Equation 15 tells us that maximum scattering occurs when the correlation length of the surface roughness is comparable to the beat length between the guided and radiation modes. For strongly guiding waveguides, where γ is small, we can approximate $f_e(x, \gamma)$ by

$$f_e(x, \gamma) \approx \frac{x\sqrt{2}}{1+x^2} \quad (16)$$

and this has a maximum at $x = 1$ of

$$f_{e,\max} = \frac{1}{\sqrt{2}} \quad (17)$$

This is, in fact, the absolute upper limit on $f_e(x, \gamma)$. It is interesting to note that now there is no longer a simple relation between the correlation length L_c and the beat length between guided and radiation modes. In this case waveguide scattering increases as the waveguide becomes more strongly guiding, in contrast to the weakly guiding situation considered above.

Combining Equation 17 with Equations 10 to 12 we see that the scattering loss α has a rigorous upper limit given by

$$\alpha \leq \frac{\sigma^2}{2k_0 d^4 n_1} g(V) \quad (18)$$

Equation 18 is an important result; for any given waveguide, independent of the fabrication details, the scattering loss has an upper limit that depends only on the magnitude of the surface roughness σ . We can easily see why there should be a maximum. If the correlation length was infinite, effectively implying no surface roughness, the scattering loss would be zero. Equally, if the correlation length was zero the surface roughness would not be resolved except at very short optical wavelengths, and scattering loss would again be zero. Consequently, α must have a maximum at some point in between. The importance of this to waveguide designers can easily be appreciated. An estimate of the deviation σ from flatness of a waveguide surface can be made much more readily than can a measurement of the statistics of the surface roughness. Equation 17 then allows an immediate estimate of the upper limit that can be expected for waveguide loss, independently of changes in the roughness statistics that may result from changes in fabrication methods.

Referring to Fig. 3, and bearing in mind that for a typical optical waveguide $\gamma \gg 1$, we can see that $f_e(x, \gamma)$ for most waveguides will change slowly as the waveguide parameters are varied, as long as x is large. If x is small, then significant changes in loss will occur as the waveguide dimensions are varied. The measurements presented in [5] suggest that x typically lies in the range 1 to 5, and that γ is about 30. Then, from Fig. 3, it would appear that the best way to reduce waveguide scattering loss for most practical waveguides, apart from reducing the

surface roughness σ , is to increase the waveguide half-width d . On the other hand, if the correlation length $L_c \gg d$, so that $x \gg 1$, then the situation is more complicated. Reducing the waveguide half-width d would decrease $f_e(x, \gamma)$, but only slowly, whereas the d^4 in the denominator of Equation 10 would cause α to increase.

3. Gaussian autocorrelation function

The Gaussian ACF is interesting, not only for its practical importance, but because a wide range of autocorrelation functions can be approximated by a Gaussian over the integration range which dominates the integrand in Equation 1. Inserting the Gaussian from Equation 5 into Equations 1 and 2 we find that the loss coefficient α can be written as [2]

$$\alpha = \varphi^2(d)(n_1^2 - n_2^2)^2 \frac{k_0^3}{4\pi n_1} \int_0^\pi \sigma^2 L_c \sqrt{\pi} \exp[-(\beta - n_2 k_0 \cos \theta)^2 L_c^2 / 4] d\theta \quad (19)$$

All the terms in Equation 19 have the same meaning as used previously. We now rewrite Equation 19, using the normalized parameters x and γ defined in Section 2, and the expression for $\varphi(d)$ in Equation 9. After some algebra we find that Equation 19 can be written in the same form as Equation 10:

$$\alpha = \frac{\sigma^2}{\sqrt{2} k_0 d^4 n_1} g(V) f_g(x, \gamma) \quad (20)$$

where $f_g(x, \gamma)$ for the Gaussian is given by

$$f_g(x, \gamma) = \frac{x}{\sqrt{2\pi}} \int_0^\pi \exp \left\{ -\frac{x^2}{4} \left[\left(1 + \frac{\gamma^2}{2} \right)^{1/2} - \frac{\gamma}{\sqrt{2}} \cos \theta \right]^2 \right\} d\theta \quad (21)$$

It is not difficult to show that the form of Equation 20 is quite general and applies to any ACF. Equation 21 cannot be computed analytically, although we shall present a very accurate approximation to it below. In Fig. 4 we present a numerical evaluation of $f_g(x, \gamma)$ as a function of x for various values of γ . If we compare this with Fig. 3 for the exponential ACF we note

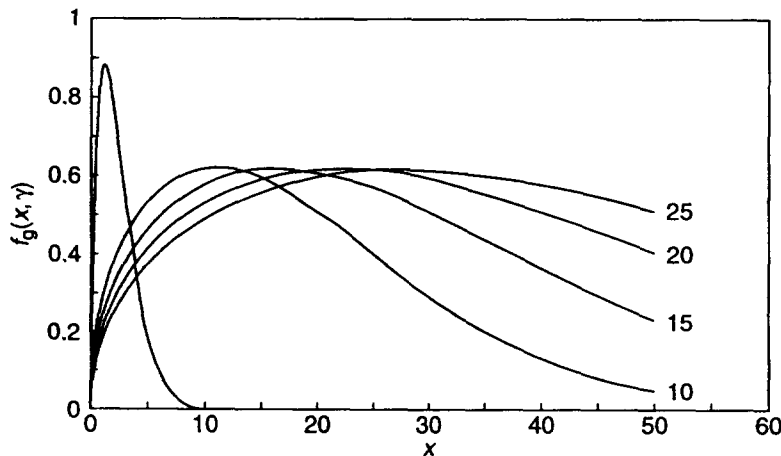


Figure 4 The same as Fig. 3 but for Gaussian statistics.

marked similarities, particularly for the important case of $\gamma \gg 1$ and $x < 10$, which as we saw in Section 2 applies to most practical waveguides. It is also clear from Fig. 4 that $f_g(x, \gamma)$ has an absolute upper bound as γ becomes small; from Equation 21 this is easily derived as

$$f_g(x, \gamma) \underset{\gamma \ll 1}{=} \sqrt{\frac{\pi}{2}} x \exp\left(-\frac{x^2}{4}\right) \quad (22)$$

Equation 22 has a maximum at $x^2 = 2$, giving the upper bound as

$$f_{g,\max} = \sqrt{\frac{\pi}{e}} \quad (23)$$

If we combine Equations 23 and 20 we find that the loss coefficient α for Gaussian statistics has an upper bound of

$$\alpha \leq \frac{\sigma^2}{k_0 d^4 n_1} g(V) \sqrt{\frac{\pi}{2e}} \quad (24)$$

A comparison of Equation 24 with Equation 18 shows that the upper limits on α for exponential or Gaussian statistics are very similar in magnitude, and reinforces the view that they represent a practical way of estimating an upper limit of the loss of any waveguide once an estimate of the magnitude of the surface roughness has been made.

Although Equation 21 cannot be evaluated analytically, we can find an extremely accurate approximation valid for weakly guiding waveguides when $\gamma \gg 1$. The calculation is based on the method of steepest descent [6] and we give the details in the appendix. We find that

$$f_g(x, \gamma) \underset{\gamma \gg 1}{\approx} 2^{-5/4} \frac{\Gamma(\frac{1}{4})}{\sqrt{\pi}} \sqrt{\frac{x}{\gamma}} \exp\left(-\frac{x^2}{16\gamma^2}\right) \left[1 - \left(\frac{x}{\gamma}\right) \frac{\Gamma(\frac{3}{4})}{\sqrt{2}\Gamma(\frac{1}{4})}\right] \quad (25)$$

where Γ is the gamma function [8], which takes the particular values

$$\Gamma(\frac{1}{4}) = 3.62560 \quad \text{and} \quad \Gamma(\frac{3}{4}) = 1.22541$$

Equation 25 is a remarkably accurate approximation, as demonstrated in Fig. 5, where it is compared to the exact numerical evaluation of Equation 21 for $\gamma = 25$. For example, at $x = 10$ the error is only 0.1%, while for $x = 50$ it is still only in error by 3%.

4. Conclusions

The main goal in this paper has been to develop analytical expressions for the scattering loss in planar optical waveguides arising as a result of irregularities in the waveguide walls. The most important results are contained in Equations 10, 18, 24 and 25, which show clearly the dependence on normalized parameters. Of particular interest for waveguide design are the expressions for upper bounds on the scattering loss, Equations 18 and 24. It is noteworthy that these expressions set very similar bounds for both exponential and Gaussian statistics. When using them it is helpful to bear in mind that for a single-mode waveguide ($V < \pi/2$) the function $g(V) < 0.95$. In general we find that, independent of the detailed statistics, the scattering loss will have an upper bound given by

$$\alpha \leq \frac{\sigma^2}{k_0 d^4 n_1} \kappa$$

where $\kappa = 0.48$ for exponential statistics, and 0.76 for Gaussian.

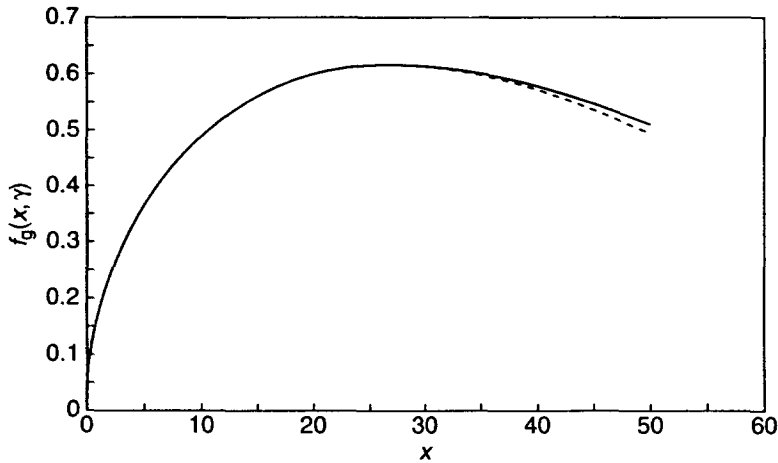


Figure 5 A comparison of $f_g(x, \gamma)$ for Gaussian statistics evaluated exactly from Equation 21 (solid line) and approximately using Equation 25 (broken line). The parameter $\gamma = 25$.

Most optical waveguides are, of course, not planar but two-dimensional. Our results can be applied to these cases by use of the effective index model to reduce them to an equivalent planar form. We also expect that the analytical methods developed in this paper will allow similar results to be derived for two-dimensional structures without the need for effective-index techniques. We hope to make this the subject of a future article.

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Appendix

Although the integral in Equation 21 cannot be evaluated exactly analytically, it is possible to obtain an excellent approximation for the weakly guiding case $\gamma \gg 1$. We use the method of steepest descent [6], which essentially involves recognizing that for large γ the integral receives most of its contribution from small values of θ . The first step therefore is to expand the integrand as a Taylor series in θ ,

$$\frac{x^2}{4} \left[\left(1 + \frac{\gamma^2}{2} \right)^{1/2} - \frac{\gamma}{\sqrt{2}} \cos \theta \right]^2 \approx \frac{x^2 \gamma^2}{8} \left(\frac{1}{\gamma^4} + \frac{\theta^2}{\gamma^2} + \frac{\theta^4}{4} - \frac{\theta^4}{12\gamma^2} \right) \quad (\text{A1})$$

where we have retained terms to fourth order in θ . For large γ and small θ we retain only the second and third terms in Equation A1, the others making an insignificant contribution to the

integral. The limits between 0 and π are also replaced by 0 and infinity, since most of the contribution to the integral comes from θ near zero. With this we can then approximate Equation 21 with

$$f_g(x, \gamma) \approx \frac{x}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{8\gamma^2}\right) \int_0^\infty \exp\left[-\left(\frac{x^2\gamma^2\theta^4}{32} + \frac{x^2\theta^2}{8}\right)\right] d\theta \quad (\text{A2})$$

The integral in Equation A2 can now be evaluated using the following result [8]:

$$\int_0^\infty \exp(-\mu z^4 - 2\nu z^2) dz = \frac{1}{4} \sqrt{\frac{2\nu}{\mu}} \exp\left(\frac{\nu^2}{2\mu}\right) K_{1/4}\left(\frac{\nu^2}{2\mu}\right) \quad (\text{A3})$$

where K_ν is a Bessel function of fractional order.

With this we obtain

$$f_g(x, \gamma) = \frac{1}{2\sqrt{2\pi}} \left(\frac{x}{\gamma}\right) \exp\left(-\frac{x^2}{16\gamma^2}\right) K_{1/4}\left(\frac{x^2}{16\gamma^2}\right) \quad (\text{A4})$$

For large γ we can now expand the argument of the Bessel function in powers of x/γ . To do this we make use of the following expansion of $K_\nu(z)$ for small z , which can be derived using standard properties of Bessel functions and which is valid for $\nu \neq 0$:

$$K_\nu(z) \approx \frac{\Gamma(\nu)}{2} \left(\frac{z}{2}\right)^{-\nu} \left[1 - \left(\frac{z}{2}\right)^{2\nu} \frac{\Gamma(1-\nu)}{\nu\Gamma(\nu)}\right] \quad (\text{A5})$$

The function $\Gamma(\nu)$ in Equation A5 is the gamma function [8]. Finally, after substituting Equation A5 into Equation A4 we obtain the result quoted in Equation 21.